

JUCBNMT at WMT2018 News Translation Task: Character Based Neural Machine Translation of Finnish to English

Sainik Kumar Mahata, Dipankar Das, Sivaji Bandyopadhyay

Computer Science and Engineering

Jadavpur University, Kolkata, India

sainik.mahata@gmail.com,

dipankar.dipnil2005@gmail.com, sivaji_cse_ju@yahoo.com

Abstract

In the current work, we present a description of the system submitted to WMT 2018 News Translation Shared task. The system was created to translate news text from Finnish to English. The system used a Character Based Neural Machine Translation model to accomplish the given task. The current paper documents the preprocessing steps, the description of the submitted system and the results produced using the same. Our system garnered a BLEU score of 12.9.

1 Introduction

Machine Translation (MT) is automated translation of one natural language to another using computer software. Translation is a tough task, not only for computers, but humans as well as it incorporates a thorough understanding of the syntax and semantics of both languages. For any MT system to return good translations, it needs good quality and sufficient amount of parallel corpus (Mahata et al., 2016, 2017).

In the modern context, MT systems can be categorized into Statistical Machine Translation (SMT) and Neural Machine Translation (NMT). SMT has had its share in making MT very popular among the masses. It includes creating statistical models, whose input parameters are derived from the analysis of bilingual text corpora, created by professional translators (Weaver, 1955). The state-of-art for SMT is Moses Toolkit¹, created by Koehn et al. (2007), incorporates subcomponents like Language Model generation, Word Alignment and Phrase Table generation. Various works have been done in SMT (Lopez, 2008; Koehn, 2009) and it has shown good results for many language pairs.

On the other hand NMT (Bahdanau et al., 2014), though relatively new, has shown considerable improvements in the translation results when compared to SMT (Mahata et al., 2018). This includes better fluency of the output and better handling of the Out-of-Vocabulary problem. Unlike SMT, it doesn't depend on alignment and phrasal unit translations (Kalchbrenner and Blunsom, 2013). On the contrary, it uses an Encoder-Decoder approach incorporating Recurrent Neural Cells (Cho et al., 2014). As a result, when given sufficient amount of training data, it gives much more accurate results when compared to SMT (Doherty et al., 2010; Vaswani et al., 2013; Liu et al., 2014).

Further, NMT can be of two types, namely Word Level NMT and Character Level NMT. Word Level NMT, though very successful, suffers from a few disadvantages. It is unable to model rare words (Lee et al., 2016). Also, since it does not learn the morphological structure of a language it suffers when accommodating morphologically rich languages (Ling et al., 2015). We can address this issue, by training the models with huge parallel corpus, but, this in turn, produces very complex and resource consuming models that aren't feasible enough.

To combat this, we plan to use Character level NMT, so that it can learn the morphological aspects of a language and construct a word, character by character, and hence tackle the rare word occurrence problem to some extent.

In the current work, we participated in the WMT 2018 News Translation Shared Task² that focused on translating news text, for European language pairs. The Character Based NMT system discussed in this paper was designed to accommodate Finnish to English translations. The orga-

¹<http://www.statmt.org/moses/>

²<http://www.statmt.org/wmt18/translation-task.html>